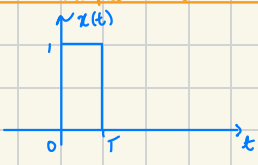
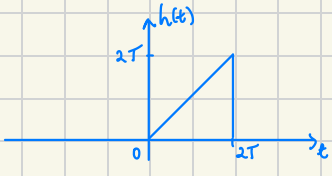
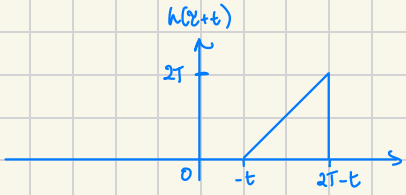
ELEC 3125 - Signals & Systems

Lecture 6 - Part 2

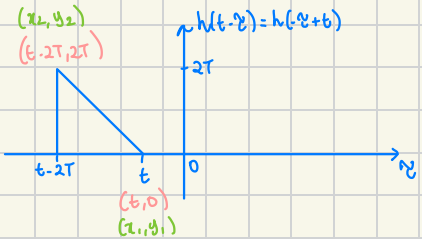
Harder example of convolution integral




$y(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)\,d\tau$, $h(t-\tau) = h(-\tau+t)$

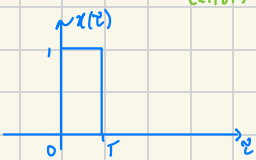



$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{2T - 0}{t - 2T - t} = -\frac{2T}{2T} = -1$ $\quad m = -1$

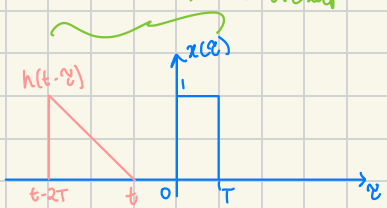
$y - y_1 = m(x - x_1) \Rightarrow y = -1(x - t) = -x + t$

In our case: $x = \tau$ & $y = h(t-\tau)$

$\boxed{h(t-\tau) = t - \tau}$




↦ no overlap

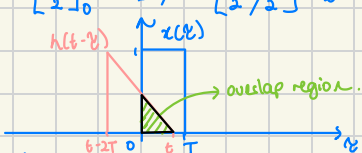



for $t \geq 0$: no overlap $\Rightarrow$ $\boxed{y(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)\,d\tau = 0, \quad t < 0}$

$0 \leq t \leq T$: $y(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)\,d\tau = \int_0^t \underbrace{x(\tau)}_{=1} h(t-\tau)\,d\tau = \int_0^t h(t-\tau)\,d\tau = \int_0^t (t-\tau)\,d\tau = \int_0^t t\,d\tau - \int_0^t \tau\,d\tau$

$= t\int_0^t d\tau - \int_0^t \tau\,d\tau = t\left[\tau\right]_0^t - \left[\frac{\tau^2}{2}\right]_0^t = t\left[t - 0\right] - \left[\frac{t^2}{2} - \frac{0^2}{2}\right] = t^2 - \frac{1}{2}t^2 = \frac{2}{2}t^2 - \frac{1}{2}t^2 = \frac{1}{2}t^2$

$\Rightarrow$ $\boxed{y(t) = \frac{1}{2}t^2, \quad 0 \leq t \leq T}$




for $t > T$ & $t - 2T \geq 0 \Rightarrow t \geq 2T \Rightarrow$ $T < t \leq 2T$

$\Rightarrow y(t) = \int_{-\infty}^{\infty} \underbrace{x(\tau)}_{=1} h(t-\tau)\,d\tau = \int_0^T (t-\tau)\,d\tau = \int_0^T t\,d\tau - \int_0^T \tau\,d\tau = t\left[\tau\right]_0^T - \left[\frac{\tau^2}{2}\right]_0^T = t\left[T - 0\right] - \left[\frac{T^2}{2} - \frac{0^2}{2}\right] = Tt - \frac{1}{2}T^2$

$\Rightarrow$ $\boxed{y(t) = Tt - \frac{1}{2}T^2, \quad T < t \leq 2T}$



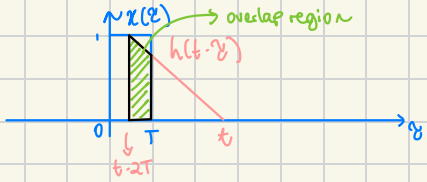

for: $t - 2T > 0$ & $t - 2T < T \Rightarrow t > 2T$ & $t < 3T \Rightarrow$ $2T < t \leq 3T$

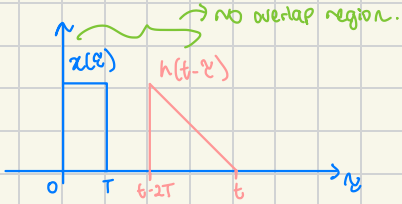
$\Rightarrow y(t) = \int_{-\infty}^{\infty} \underbrace{x(\tau)}_{=1} h(t-\tau)\,d\tau = \int_{t-2T}^{T} (t-\tau)\,d\tau = \int_{t-2T}^{T} t\,d\tau - \int_{t-2T}^{T} \tau\,d\tau = t\left[\tau\right]_{t-2T}^{T} - \left[\frac{\tau^2}{2}\right]_{t-2T}^{T} = t\left[T - (t-2T)\right] - \left[\frac{(T)^2}{2} - \frac{(t-2T)^2}{2}\right]$

$= t\left[T - t + 2T\right] - \left[\frac{T^2}{2} - \frac{(t^2 - 2tT - 2tT + 4T^2)}{2}\right] = t\left[3T - t\right] - \left[\frac{T^2}{2} - \frac{(t^2 - 4tT + 4T^2)}{2}\right] = 3tT - t^2 - \left[\frac{T^2}{2} - \frac{t^2}{2} + 2tT - 2T^2\right]$

$= 3tT - t^2 - \left[\frac{T^2}{2} - \frac{t^2}{2} + 2tT - \frac{4T^2}{2}\right] = 3tT - t^2 - \left[-\frac{3}{2}T^2 - \frac{t^2}{2} + 2tT\right] = 3tT - t^2 + \frac{3}{2}T^2 + \frac{t^2}{2} - 2tT = -\frac{1}{2}t^2 + Tt + \frac{3}{2}T^2$

→ $y(t) = -\frac{1}{2}t^2 + Tt + \frac{3}{2}T^2$ , $2T < t \leq 3T$




for $t > 3T$ : $y(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau = 0$

Overall output

$$y(t) = \begin{cases} 0 & , t < 0 \\ \frac{1}{2}t^2 & , 0 < t < T \\ Tt - \frac{1}{2}T^2 & , T < t < 2T \\ -\frac{1}{2}t^2 + Tt + \frac{3}{2}T^2 & , 2T < t < 3T \\ 0 & , t > 3T \end{cases}$$

Overall graph

